Algebraic Geometry

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Contents

0.1 Plan . . . 1
0.2 Resources . . . 1
0.3 Why scheme theory? . . . 2

1 Beyond varieties **3**
1.1 Summary of classical algebraic geometry . . . 3
1.2 Limitations . . . 4
1.3 Spectrum of a ring . . . 4
1.4 Topology . . . 5
1.5 Functions on opens . . . 6

2 Sheaves **8**
2.1 Presheaves . . . 8
2.2 Sheaves . . . 9
2.3 Basic constructions . . . 10
2.4 Kernels, Cokernels, etc. . . . 12
2.5 Moving between spaces . . . 13

3 Schemes **15**
3.1 Sheaf on a base . . . 15
3.2 Examples of schemes . . . 17

Lecture 1

0.1 Plan

1. Basics of sheaves of topological spaces
2. Definitions of schemes and of morphisms
3. Properties of schemes (the algebraic geometry analogues of compactness)
4. Rapid introduction to the cohomology of sheaves

0.2 Resources

1. Course webpage - Google Dhruv
2. Books:
   - Hartshorne - Algebraic Geometry, Chapter 2 and $\varepsilon\times$Chapter 3
   - Vakil - Rising Sea, good online (free) alternative
   - Eisenbud-Harris - Geometry of schemes
3. Four examples sheets on webpage
4. SAGES - Look on Dhruv's webpage

*Based on lectures by Dhruv Ranganathan. Last updated October 23, 2023.

0.3 Why scheme theory?

Moduli theory – It is better to study families of varieties than one at a time. Or better, all varieties of a given type at the same time.

Examples of moduli: The set of all lines in $\mathbb{P}^2$. A line in $\mathbb{P}^2$ is

$$\{aX_0 + bX_1 + cX_2 = 0\}$$

In particular, as lines are paramterised by the triple (a, b, c), we have a correspondence

$$\mathbb{P}^2_{\text{dual}} = \text{lines in } \mathbb{P}^2 \leftrightarrow \mathbb{P}^2$$

The same logic applies for degree d hypersurface in $\mathbb{P}^n$, and we get

$$\text{degree } d \text{ hypersufaces in } \mathbb{P}^n \leftrightarrow \mathbb{P}^N$$

where $N = \binom{n+d}{d} - 1$. However, there is something wrong with this picture. Some polynomials are of the form

$$f = f_1^2 f_2$$

But in this case,

$$\mathbb{V}(f) = \mathbb{V}(f_1 f_2)$$

and $f_1 f_2$ is (in general) not of degree d.

A solution Take $U_d \subseteq \mathbb{P}^N$, where

$$[f] \in U_d \iff f \text{ has no repeated factors}$$

But U_d is not compact.

Output of scheme theory for a fixed projective space $\mathbb{P}^n$, we obtain a space

$$\text{subvarieties of } \mathbb{P}^n \leftrightarrow \text{Var}(\mathbb{P}^n) \subsetneq \text{Hilb}(\mathbb{P}^n) \leftrightarrow \text{ subschemes of } \mathbb{P}^n$$

where $\text{Hilb}(\mathbb{P}^n)$ is the Hilbert scheme of $\mathbb{P}^n$, which is compact in the Euclidean topology. That is, the limit of varieties need not be a variety, but limits of schemes are always schemes.

In scheme theory,

$$\mathbb{V}(X_0 + X_1 + X_2) \text{ and } \mathbb{V}((X_0 + X_1 + X_2)^2)$$

are not isomorphic as schemes.

Weil conjectures Fix $f \in \mathbb{Z}[X_0, \ldots, X_{n+1}]$ homogeneous. We have

$$X = \mathbb{V}(f) \subseteq \mathbb{P}^{n+1}_{\mathbb{C}}$$

We will assume that X is smooth. That is, X is a (complex) manifold. In particular, X is a compact topological space, and so we have numbers $b_0(X), \ldots, b_{2n}(X)$ called the Betti numbers, where

$$b_i(X) = \text{rank}(H_i(X, \mathbb{Z}))$$

In particular, we have the Euler characteristic $\chi(X)$.

On the other hand, fix a prime number p, and let

$$N_m = \text{number of solutions to } f \text{ in } \mathbb{F}_{p^m} = \left|X(\mathbb{F}_{p^m})\right|$$

Define the Weil ζ function

$$\zeta(X; t) = \exp\left(\sum_m \frac{N_m}{m} t^m\right)$$

Theorem 0.3.1 (Weil conjectures, Grothendieck).

1. $\zeta(X; t)$ is a rational function in t
2. Moreover,
$$\zeta(X; t) = \frac{P_0(t)P_2(t)\cdots P_{2n}(t)}{P_1(t)P_3(t)\cdots P_{2n-1}(t)}$$
where $\deg(P_i(t)) = b_i(X)$.

Upshot: The proof is (fundamentally) via scheme theory. We need a space $\mathfrak{X}$ which interpolates geometry over $\mathbb{C}$ and geometry over finite fields.

Lecture 2

1 Beyond varieties

1.1 Summary of classical algebraic geometry

Let $k = \overline{k}$ be an algebraically closed field. Define the *affine space*

$$\mathbb{A}_k^n = k^n$$

as a set.

An *affine variety* is a subset

$$V = \mathbb{V}(S) \subseteq \mathbb{A}_k^n$$

for some $S \subseteq k[x_1, \ldots, x_n]$. Note that $\mathbb{V}(S) = \mathbb{V}(\langle S \rangle)$, where $\langle S \rangle$ is the ideal generated by S. By the Hilbert basis theorem, or equivalently, $k[x_1, \ldots, x_n]$ is Noetherian, $\langle S \rangle$ is generated by finitely many elements. Moreover,

$$\mathbb{V}(I) = \mathbb{V}\left(\sqrt{I}\right)$$

where I is the *radical* of I, given by

$$\sqrt{I} = \{f \mid f^m \in I \text{ for some } m \in \mathbb{N}\}$$

Given varieties $V \subseteq \mathbb{A}_k^n$, $W \subseteq \mathbb{A}_k^m$, a *morphism* $\varphi : V \to W$ is a function $V \to W$, such that if we write $\varphi = (f_1, \ldots, f_m)$, each f_i is a restriction of a polynomial in $k[x_1, \ldots, x_n]$. An *isomorphism* $\varphi : V \to W$ is a morphism, such that $\psi : W \to V$ also a morphism, $\psi \circ \varphi = \mathrm{id}_V$, $\varphi \circ \psi = \mathrm{id}_W$.

The basic correspondence is

$$\frac{\text{affine varieties over } k}{\text{isomorphism}} \leftrightarrow \text{finitely generated } k\text{-algebras } A \text{ without nilpotent elements}$$

How? Given a variety V (representing an isomorphism class of such), then we can write $V = \mathbb{V}(I)$. Moreover, we can assume that I is a radical ideal. We map

$$V \mapsto \frac{k[x_1, \ldots, x_n]}{I}$$

Conversely, if A is a finitely generated nilpotent-free k-algebra, then by definition,

$$A \cong \frac{k[y_1, \ldots, y_n]}{J}$$

where J is a radical ideal.

Note that we need to check that both maps are well-defined.

Notation 1.1.1. The algebra associated to V is (classically) denoted $k[V]$, and called the *coordinate ring of* V.

Compatibility of morphisms

Note that we have a correspondence

$$\{\text{morphisms } V \to W\} \leftrightarrow \{k\text{-algebra homomorphisms } k[W] \to k[V]\}$$

Zariski topology

Suppose $V = \mathbb{V}(I) \subseteq \mathbb{A}_k^n$ is an affine variety, with coordinate ring $k[V]$. The *Zariski topology* has closed sets

$$V \cap \mathbb{V}(S)$$

where $S \subseteq k[x_1, \ldots, x_n]$.

If $V \cong W$ as varieties, then the topological spaces with the Zariski topology are also homeomorphic.

Nullstellensatz

Fix a variety V, and let $k[V]$ be its coordinate ring. Given a point $p \in V$, we can produce a homomorphism

$$\begin{aligned} \mathrm{ev}_p : k[V] &\to k \\ \mathrm{ev}_p(f) &= f(p) \end{aligned}$$

Moreover, ev_p is surjective, for example, by taking the constant functions. With this,

$$\mathfrak{m}_p = \ker(\mathrm{ev}_p) \trianglelefteq k[V]$$

is a maximal ideal. So we get a map

$$\{\text{points } p \in V\} \to \{\text{maximal ideals in } k[V]\}$$

Hilbert's Nullstellensatz $\implies$ the above map is a bijection.
Therefore, points of V and maximal ideals in $k[V]$ are "the same thing".

1.2 Limitations

What is an abstract variety?

That is, what is a topological space X, such that we have an open cover $\{U_i\}$, where each U_i is an affine variety, which is compatible on overlaps? For example, we have projective space $\mathbb{P}^n$.

Example 1.2.1 (non-algebraically closed fields)
Take the ideal

$$I = \left\langle x^2 + y^2 + 1 \right\rangle \trianglelefteq \mathbb{R}[x, y]$$

Observe $\mathbb{V}(I) = \varnothing \subseteq \mathbb{R}^2$, but I is prime, so radical. Hence the Nullstellensatz fails in this case.

On what topological space is $R[x, y]/\left\langle x^2 + y^2 + 1 \right\rangle$ "naturally" the set of functions?

Or more generally, $\mathbb{Z}, \mathbb{Z}[x]$ and so on.

Example 1.2.2 (why restrict to radical ideals, or nilpotent free algebras)
Let $C = \mathbb{V}(y - x^2) \subseteq \mathbb{A}^2_k$, and $D = \mathbb{V}(y) \subseteq \mathbb{A}^2_k$. Then

$$C \cap D = \mathbb{V}(x, y) = \{(0, 0)\}$$

If we set $D_\delta = \mathbb{V}(y - \delta)$, then $C \cap D_\delta$ is two points, for all $\delta \neq 0$.

Lecture 3

1.3 Spectrum of a ring

Let A be a (commutative, unital) ring.

Definition 1.3.1 (Zariski spectrum)
The *Zariski spectrum of* A is

$$\operatorname{Spec} A = \{\mathfrak{p} \trianglelefteq A \mid \mathfrak{p} \text{ is a prime ideal}\}$$

Given a ring homomorphism $\varphi : A \to B$, we have a induced map (of sets)

$$\begin{aligned} \varphi^{-1} : \operatorname{Spec}(B) &\to \operatorname{Spec} A \\ \mathfrak{q} &\mapsto \varphi^{-1}(\mathfrak{q}) \end{aligned}$$

Warning: This would fail if we considered maximal ideals instead. That is, the preimage of a maximal ideal need not be maximal.

Given $f \in A$ and $\mathfrak{p} \in \operatorname{Spec} A$, we have an induced

$$\overline{f} \in A/\mathfrak{p}$$

by taking the quotient. Informally, we can evaluate any $f \in A$ at points $\mathfrak{p} \in \operatorname{Spec}(A)$, with the caviat that the codomain of the evaluation map depends on $\mathfrak{p}$.

Example 1.3.2

Let $A = \mathbb{Z}$, then

$$\operatorname{Spec}(\mathbb{Z}) = \{p \mid p \text{ a prime number}\} \cup \{0\}$$

Choose an element, say $132 \in \mathbb{Z}$. Given a prime p, we can look at $132 \mod p \in \mathbb{Z}/p$.

Takeaway:

$$\operatorname{Spec}(\mathbb{Z}) \rightsquigarrow \text{space}$$

$$132 \in \mathbb{Z} \rightsquigarrow \text{a function}$$

$$132 \mod p \rightsquigarrow \text{value of the function at } p$$

Example 1.3.3

$A = \mathbb{R}[x]$, then

$$\operatorname{Spec} \mathbb{R}[x] = \frac{\mathbb{C}}{\text{complex conjugation}} \cup \{0\}$$

Exercise: Draw $\operatorname{Spec} \mathbb{Z}[x]$ and $\operatorname{Spec} k[x]$ for any field k.

Example 1.3.4 (for sanity)

If $A = \mathbb{C}[x]$, then

$$\operatorname{Spec} A = \mathbb{C} \cup \{0\}$$

Given $a \in \mathbb{C}$, we have the maximal ideal $\langle z - a \rangle$.

1.4 Topology

Fix $f \in A$, then define

$$\mathbb{V}(f) = \{\mathfrak{p} \in \operatorname{Spec}(A) \mid f \equiv 0 \pmod{\mathfrak{p}}\} \subseteq \operatorname{Spec}(A)$$

or equivalently, $f + \mathfrak{p} = 0 \in A/\mathfrak{p}$, or $f \in \mathfrak{p}$. Similarly, for $J \trianglelefteq A$ an ideal,

$$\mathbb{V}(J) = \{\mathfrak{p} \in \operatorname{Spec}(A) \mid J \subseteq \mathfrak{p}\}$$

Proposition 1.4.1. The sets $\mathbb{V}(J) \subseteq \operatorname{Spec} A$ ranging over all ideals $J \trianglelefteq A$, form the closed sets of a topology on $\operatorname{Spec} A$. This topology is called the *Zariski topology*.

Proof. Easy facts that $\varnothing = \mathbb{V}(1)$ and $\operatorname{Spec} A = \mathbb{V}(0)$ are closed. Since

$$\mathbb{V}\left(\sum_\alpha I_\alpha\right) = \bigcap_\alpha \mathbb{V}(I_\alpha)$$

the intersections of closed sets is closed. Finally,

$$\mathbb{V}(I_1 \cap I_2) = \mathbb{V}(I_1) \cup \mathbb{V}(I_2)$$

$\supseteq$ is clear, conversely, $I_1 I_2 \subseteq I_1 \cap I_2 \subseteq \mathfrak{p}$, then by primality of $\mathfrak{p}$, $I_1 \subseteq \mathfrak{p}$ or $I_2 \subseteq \mathfrak{p}$. □

Example 1.4.2

Consider $\operatorname{Spec}\mathbb{C}[x,y]$. A few observations:

- $0 \in \operatorname{Spec}\mathbb{C}[x,y]$ is dense in the Zariski topology. That is,
$$\overline{\{\langle 0\rangle\}} = \operatorname{Spec}\mathbb{C}[x,y]$$
since every prime ideal contains 0.
- Consider the prime ideal $\langle y^2 - x^3\rangle$. Consider a maximal ideal
$$\mathfrak{m}_{a,b} = \langle x-a, y-b\rangle$$
When is $\mathfrak{m}_{a,b} \in \overline{\{\langle y^2-x^3\rangle\}}$? This holds if and only if $b^2 = a^3$. See examples sheet 1.

Remark 1.4.3. Points are not closed.

1.5 Functions on opens

Let $f \in A$. Define the *distinguished open corresponding to* f to be
$$U_f = \operatorname{Spec}(A) \setminus \mathbb{V}(f)$$

Example 1.5.1

If $A = \mathbb{C}[x]$, then
$$\operatorname{Spec} A = \mathbb{C} \cup \{\langle 0\rangle\}$$
Take $f = x$. Recall that we have a bijection
$$\begin{aligned}\operatorname{Spec}(A) &\leftrightarrow \mathbb{C} \cup \{\langle 0\rangle\}\\ (x-a) &\mapsto a \in \mathbb{C}\\ 0 &\mapsto 0\end{aligned}$$
Then $\mathbb{V}(x) = \{\mathfrak{p} \in \operatorname{Spec} A \mid x \in \mathfrak{p}\} = \{\langle x\rangle\}$. So
$$U_x = \operatorname{Spec}(A) \setminus \{\langle 0\rangle\}$$
As a picture, $\operatorname{Spec}(\mathbb{C}[x])$ is a line (corresponding to $\mathbb{C}$) with a "generic point" $\{0\}$.

More generally, suppose we fix $a_1, \dots, a_r \in \mathbb{C}$, with
$$U = \operatorname{Spec}(A) \setminus \{\langle x-a_i\rangle\}_{i=1}^r$$
then $U = U_f$ where $f = \prod_{i=1}^r (x-a_i)$.

Lemma 1.5.2. The distinguished opens U_f for all $f \in A$ form a basis for the Zariski topology on $\operatorname{Spec} A$.

Proof. Examples sheet 1. □

A bit of commutative algebra: Given $f \in A$, the *localisation of* A *at* f is
$$A_f = \frac{A[x]}{\langle xf - 1\rangle} = A\left[\frac{1}{f}\right]$$

Lemma 1.5.3. The distinguished open $U_f \subseteq \operatorname{Spec}(A)$ is naturally homeomorphic to $\operatorname{Spec}(A_f)$, via the ring homomorphism

$$j : A \to A_f$$

which induces

$$j^{-1} : \operatorname{Spec}(A_f) \to \operatorname{Spec} A$$

Proof. Primes in A_f are in bijection with primes of A, which miss f, via j^{-1}. To see this, the preimage of a prime ideal is prime, and so if $\mathfrak{q} \trianglelefteq A_f$ is prime, then $j^{-1}(\mathfrak{a}) \trianglelefteq A$ is prime. For the converse, given $\mathfrak{p} \trianglelefteq A$, define

$$\mathfrak{p}_f = j(\mathfrak{p})A_f = \langle j(\mathfrak{p}) \rangle$$

Claim 1.5.4. $\mathfrak{p}_f$ is a prime exactly when $f \notin \mathfrak{p}$.

Proof of claim. To see this, f is a unit in A_f, and so if $f \in \mathfrak{p}$, then $\mathfrak{p}_f$ contains a unit, and so $\mathfrak{p}_f = 1$.

But if $f \notin \mathfrak{p}$, then

$$\frac{A_f}{\mathfrak{p}_f} \cong \left(\frac{A}{\mathfrak{p}}\right)_{\overline{f}}$$

where $\overline{f} = f + \mathfrak{p}$. So

$$(A/\mathfrak{p})_{\overline{f}} \subseteq \operatorname{Frac}(A/\mathfrak{p})$$

and so it is an integral domain. □

Finally, we can check that this defines a bijection, but this is just basic properties of ideals. □

Facts about distinguised opens

- $U_f \cap U_g = U_{fg}$.
- $U_{f^n} = U_f$ for all $n \geq 1$.
- The rings A_f and A_{f^n} (for $n \geq 1$) are isomorphic. To see this,

  $$A_f = \frac{A[x]}{\langle xf - 1 \rangle} \quad \text{and} \quad A_{f^n} = \frac{A[y]}{\langle yf^n - 1 \rangle}$$

  The isomorphism is given by

  $$y \mapsto x^n \quad \text{and} \quad x \mapsto f^{n-1}y$$

  which define inverse maps.
- Containment: $U_f \subseteq U_g$ if and only if f^n is a multiple of g for some $n \geq 1$. (Recall if $f = gf'$, then certainly $U_f \subseteq U_g$)

  Proof. The "if" direction is clear. Conversely, if $U_f \subseteq U_g$. That is,

  $$\mathbb{V}(f) \supseteq \mathbb{V}(g)$$

  The set $\mathbb{V}(f)$, by definition, is the set of all primes, containing f.

  Claim 1.5.5. $\sqrt{\langle f \rangle} \subseteq \sqrt{\langle g \rangle}$.

  To see this, the radical of an ideal I is the intersection of all primes containing I, which we will see in Commutative Algebra. □

Foreshadowing

Fix A, we've made an assignment

$$\begin{aligned} \text{distinguised opens in } \operatorname{Spec} A &\to \text{rings} \\ U_f &\mapsto A_f \end{aligned}$$

The association is functorial, that is, if $U_{f_1} \subseteq U_{f_2}$, then we can assume $f_1^n = f_2 f_3$, so $U_{f_1} = U_{f_1^n} = U_{f_2 f_3} \subseteq U_{f_2}$, and so there is a homomorphism

$$A_{f_2} \to A_{f_1}$$

which is the restriction map.

Can we extend this association to all open sets? See notes.

2 Sheaves

2.1 Presheaves

Let X be a topological space.

Definition 2.1.1 (presheaf)

A *presheaf!of abelian groups* is an association

$$\begin{aligned} \{\text{Open set in } X\} &\leftrightarrow \text{Abelian Groups} \\ U &\mapsto \mathcal{F}(U) \end{aligned}$$

and for $U \subseteq V$ opens, we have a homomorphism

$$\operatorname{res}_U^V : \mathcal{F}(V) \to \mathcal{F}(U)$$

called the *restriction map*, such that

1. $\operatorname{res}_U^U = \mathrm{id}$,
2. $\operatorname{res}_U^V \operatorname{res}_V^W = \operatorname{res}_U^W$ for $U \subseteq V \subseteq W$.

Example 2.1.2

For any space X, take

$$\mathcal{F}(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous}\}$$

with $\operatorname{res}_U^V(f) = f|_U$.

We can make an analogous definition for sheaves of rings, sets, R-modules, and so on.

Definition 2.1.3 (morphism of presheaves)

A *morphism* $\varphi : \mathcal{F} \to \mathcal{G}$ *of presheaves* on X is, for each $U \subseteq X$ open, a homomorphism

$$\varphi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$$

compatible with restrictions, that is, the diagram

$$\begin{array}{ccc} \mathcal{F}(V) & \xrightarrow{\varphi(V)} & \mathcal{G}(V) \\ \downarrow{\scriptstyle \operatorname{res}_U^V} & & \downarrow{\scriptstyle \operatorname{res}_U^V} \\ \mathcal{F}(U) & \xrightarrow[\varphi(U)]{} & \mathcal{G}(U) \end{array}$$

commutes, for all $U \subseteq V$.

In terms of category theory, the opens in X form a category, with morphisms given by inclusions. In this case, a presheaf is a contravariant functor $\mathrm{Opens}(X) \to \mathrm{Ab}$, and a morphism is a natural transformation.

A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ is *injective* (resp. *surjective*) if $\varphi(U) : \mathcal{F}(U) \to \mathcal{G}(U)$ is injective (resp. surjective) for all $U \subseteq X$.

Lecture 5

2.2 Sheaves

Definition 2.2.1 (sheaf)

A *sheaf* is a presheaf $\mathcal{F}$ on X, such that

S1. Supppose $U \subseteq X$ open, $\{U_i\}$ an open cover of U, then for $s \in \mathcal{F}(U)$, with

$$s|_{U_i} = \mathrm{res}^U_{U_i} s = 0$$

for all i, then $s = 0$.

S2. Suppose $U \subseteq X$ is open, $\{U_i\}$ an open cover of U, given $s_i \in \mathcal{F}(U_i)$, with

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

then there exists $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$.

Remark 2.2.2. These axioms imply that $\mathcal{F}(\varnothing) = 0$.

Definition 2.2.3 (morphism of sheaves)

A *morphisms of sheaves* $\varphi : \mathcal{F} \to \mathcal{G}$ is just a morphism of the underlying presheaves.

Example 2.2.4

If X is a topological space, and

$$\mathcal{F}(U) = \{f : U \to \mathbb{R} \text{ continuous}\}$$

Then $\mathcal{F}$ is a sheaf.

Example 2.2.5 (non-example)

Let $X = \mathbb{C}$ with the usual Euclidean topology. Let

$$\mathcal{F}(U) = \{f : U \to \mathbb{C} \text{ holomorphic and bounded}\}$$

The issue here is that function being bounded is not a local property. That is, we can glue bounded functions together and get an unbounded function.

Example 2.2.6 (non-example)

Fix an abelian group G, and set

$$\mathcal{F}(U) = G$$

This is called the *constant presheaf*. If U_1, U_2 are disjoint, then

$$\mathcal{F}(U_1 \cup U_2) = \mathcal{F}(U_1) \oplus \mathcal{F}(U_2) = G \oplus G \neq G$$

(unless $G = 0$ that is).

Example 2.2.7 (constant sheaf)
Let G be an abelian group, with the discrete topology. Then define

$$\mathcal{F}(U) = \{f : U \to G \text{ continuous}\} = \{f : U \to G \text{ locally constant}\}$$

This is called the *constant sheaf*.

Example 2.2.8
If V is an irreducible variety, define

$$\mathcal{O}_V(U) = \{f \in k(V) \mid f \text{ is regular at all } p \in U\}$$

Here, regular at p means $f = g/h$, $g, h \in k[V]$, with $h(p) \neq 0$. $\mathcal{O}_X$ is called the *structure sheaf* of V. This is a sheaf as being regular is a local condition.

2.3 Basic constructions

Definition 2.3.1 (section)
Let $\mathcal{F}$ be a sheaf, a *section* s of $\mathcal{F}$ on U is an element $s \in \mathcal{F}(U)$.

Definition 2.3.2 (Stalks)
Fix $p \in X$, and $\mathcal{F}$ a presheaf on X. The *stalk of* $\mathcal{F}$ *at* p is

$$\mathcal{F}_p = \frac{\{(U, s) \mid U \text{ open neighbourhood of } p, s \in \mathcal{F}(U)\}}{\sim}$$

where $(U, s) \sim (V, t)$ if there exists $W \subseteq U \cap V$ an open neighbourhood of p, such that

$$s|_W = t|_W$$

Elements of $\mathcal{F}_p$ are called *germs*.

Example 2.3.3
Consider the affine line $\mathbb{A}^1$, then

$$\mathcal{O}_{\mathbb{A}^1,0} = \left\{ \frac{f(t)}{g(t)} \;\middle|\; g(0) \neq 0 \right\} \subseteq k(t)$$

Equivalently, we can consider this as the localisation $k[t]_{(t)}$.

Proposition 2.3.4. Suppose $f : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves on X, such that for all $p \in X$, the induced map

$$f_p : \mathcal{F}_p \to \mathcal{G}_p$$

is an isomorphism. Then f is an isomorphism.

Note that

$$f_p((U, s)) = (U, f_U(s))$$

which is well defined by definition of a morphism of sheaves.

Proof. We will show that

$$f_U : \mathcal{F}(U) \to \mathcal{G}(U)$$

is an isomorphism for all U. Once we know this, we will define f^{-1} by

$$(f^{-1})_U = f_U^{-1}$$

f_U **is injective.** Suppose $s \in \mathcal{F}(U)$ has $f_U(s) = 0$. By injectivity of f_p, $(U, s) = 0$ in $\mathcal{F}_p$ for all $p \in U$. This means that for all $p \in U$, there exists an open neighbourhood U_p of p such that $s|_{U_p} = 0$. This defines a cover of U. Hence by S1, $s = 0$.

f_U **is surjective.** Given $t \in \mathcal{G}(U)$, for each $p \in U$, we have $(U_p, s_p) \in \mathcal{F}_p$, with $f_p(U_p, s_p) = (U, t) \in \mathcal{G}_p$. By shrinking U_p if necessary, we can assume $f_{U_p}(s_p) = t|_{U_p}$. For points $p, q \in U$,

$$f_{U_p \cap U_q}(s_p|_{U_p \cap U_q} - s_q|_{U_p \cap U_q}) = t|_{U_p \cap U_q} - t|_{U_p \cap U_q} = 0$$

Thus bu injectivity of $f_{U_p \cap U_q}$,

$$s_p|_{U_p \cap U_q} = s_q|_{U_p \cap U_q}$$

Thus by S2, there exists $s \in \mathcal{F}(U)$, with $s|_{U_p} = s_p$. Now

$$f_U(s)|_{U_p} = f_U(s|_{U_p}) = f|_{U_p}(s_p) = t|_{U_p}$$

Thus, $f_U(s) = t$ by S1. □

Remark 2.3.5. There is an asymmetry in the proof here, we need to show injectivity before surjectivity.

Lecture 6

Exercises:

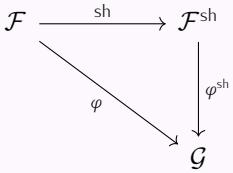
1. There is a map

$$\mathcal{F}(U) \to \prod_{p \in U} \mathcal{F}_p$$
$$s \mapsto ((U, s))_{p \in U}$$

2. Given morphisms $\varphi, \psi : \mathcal{F} \to \mathcal{G}$, with $\varphi_p = \psi_p$ for all $p \in X$, we have that $\varphi = \psi$.

Definition 2.3.6 (sheafification)

Suppose $\mathcal{F}$ is a presheaf on X, then a morphism $\mathrm{sh} : \mathcal{F} \to \mathcal{F}^{\mathrm{sh}}$, where $\mathcal{F}^{\mathrm{sh}}$ is a sheaf, is a *sheafiifcation* if for any morphism $\varphi : \mathcal{F} \to \mathcal{G}$, where $\mathcal{G}$ is a sheaf, there exists a unique morphism $f^{\mathrm{sh}} : \mathcal{F}^{\mathrm{sh}} \to \mathcal{G}$, making the diagram



commute.

Remark 2.3.7. (i) We have defined sheafification by a universal property, and so as usual, it is unique up to unique isomorphism.

(ii) A morphism on presheaves $\mathcal{F} \to \mathcal{G}$ induces a morphism of sheaves $\mathcal{F}^{\mathrm{sh}} \to \mathcal{G}^{\mathrm{sh}}$.

Proposition 2.3.8. Sheafification exists.

Construction. Given a presheaf $\mathcal{F}$ on X, define

$$\mathcal{F}^{\text{sh}}(U) = \left\{ f : U \to \bigsqcup_{p \in U} \mathcal{F}_p \;\middle|\; (*) \right\}$$

where the condition $(*)$ is that: $f(p) \in \mathcal{F}_p$ and for every $p \in U$ there exists an open neighbourhood $V_p \subseteq U$ of p, and $s \in \mathcal{F}(V_p)$ such that $(V_p, s) = f(q) \in \mathcal{F}_q$ for all $q \in V_p$.

It is clear that this is a sheaf, and we leave as an exercise to check that this satisfies the universal property. □

Corollary 2.3.9. The stalks of $\mathcal{F}$ and $\mathcal{F}^{\text{sh}}$ coincide.

Proof. From the definition of sheafification. □

Exercise: Find a non-zero presheaf $\mathcal{F}$ with $\mathcal{F}^{\text{sh}} = 0$.

2.4 Kernels, Cokernels, etc.

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. Then we can define presheaves $\ker(\varphi), \text{im}(\varphi), \text{coker}(\varphi)$, where on an open set U, we define

$$\begin{aligned} \ker(\varphi)(U) &= \ker(\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)) \\ \text{coker}(\varphi)(U) &= \text{coker}(\varphi_U) \\ \text{im}(\varphi)(U) &= \text{im}(\varphi_U) \end{aligned}$$

These are all presheaves.

Exercise: The presheaf kernel for a morphism $\varphi : \mathcal{F} \to \mathcal{G}$ between sheaves is a sheaf.

However, in general, $\text{coker}(\varphi)$ is not a sheaf.

Example 2.4.1

Let $X = \mathbb{C}$ with the usual Euclidean topology, and let

$$\mathcal{O}_X(U) = \{f : U \to \mathbb{C} \text{ holomorphic}\}$$

with addition, and set

$$\mathcal{O}_X^*(U) = \{f : U \to \mathbb{C} \text{ holomorphic nowhere vanishing}\}$$

with multiplication. We have a morphism of sheaves $\exp : \mathcal{O}_X \to \mathcal{O}_X^*$, given by

$$\exp(f)(z) = \exp(f(z))$$

The kernel of exp is $2\pi i\mathbb{Z}$, where $\mathbb{Z}$ is the constant sheaf. But the cokernel is not a sheaf. Let $U_1 = \mathbb{C}\backslash[0, \infty)$ and $U_2 = \mathbb{C} \setminus (-\infty, 0]$. Let $U = U_1 \cup U_2 = \mathbb{C} \setminus 0$.

Let $f(z) = z$. Then $f \in \mathcal{O}_x^*(U)$, and is not in the image of $\exp : \mathcal{O}_X(U) \to \mathcal{O}_X^*(U)$, since $\log(z)$ is not defined on U. Thus f defines a non-zero section of $\text{coker}(\exp)(U)$. But if we restrict to U_i, then f is in the image of exp. With this, $f|_{U_i} = 1 \in \text{coker}(\exp)(U_i)$. Hence coker(exp) does not satisfy S1.

Definition 2.4.2 (sheaf image, sheaf cokernel)

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. We define the *sheaf cokernel* and the *sheaf image* of φ to be $\text{coker}(\varphi)^{\text{sh}}$ and $\text{im}(\varphi)^{\text{sh}}$ respectively.

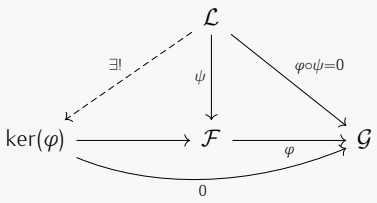
We say that a morphism $f : \mathcal{F} \to \mathcal{G}$ is *injective* if $\ker(f) = 0$, and *surjective* if $\mathrm{im}(f) = \mathcal{G}$.
Warning: $f : \mathcal{F} \to \mathcal{G}$ being surjective does *not* imply that the map $f_U : \mathcal{F}(U) \to \mathcal{G}(U)$ is surjective for all U[1].

Remark 2.4.3. A crucial fact is that there exists an exact sequence

$$0 \longrightarrow 2\pi i\mathbb{Z} \longrightarrow \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^* \longrightarrow 1$$

That is, $2\pi i\mathbb{Z}$ is the kernel of exp, and $\mathrm{coker}(\exp) = 1$.

Remark 2.4.4. $\ker(\varphi)$ and $\mathrm{coker}(\varphi)$ satisfies the category theoretic definitions (i.e. in abelian categories).
Suppose $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism, then $\ker(\varphi)$ is the object such that given $\psi : \mathcal{L} \to \mathcal{F}$, with $\varphi \circ \psi = 0$, then ψ factors through $\ker(\varphi)$.



It is easy to check this by working locally, that is, in open sets. There is an analogous definition for the cokernel.
With this, we see that the sheaf of abelian groups is an abelian category.

Remark 2.4.5. 1. We can also define subsheafs, $\mathcal{F} \subseteq \mathcal{G}$ if there exists inclusions $\mathcal{F}(U) \subseteq \mathcal{G}(U)$ compatible with restrictions. For example, $\ker(\varphi : \mathcal{F} \to \mathcal{G})$ is a subsheaf of $\mathcal{F}$.

Lecture 7

2.5 Moving between spaces

Given $f : X \to Y$ continuous, with sheaves $\mathcal{F}$ on X and $\mathcal{G}$ on Y.

Definition 2.5.1 (pushforward, direct image)
Define the *presheaf pushforward* $f_*\mathcal{F}$ by

$$f_*\mathcal{F}(U) = \mathcal{F}(f^{-1}(U))$$

for an open $U \subseteq Y$.

Proposition 2.5.2. The presheaf pushforward of a sheaf is a sheaf.

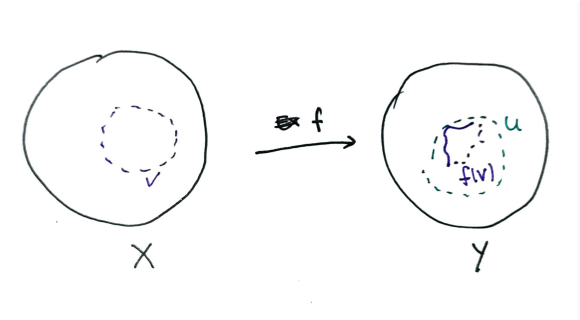
Proof. Trivial. □

Definition 2.5.3 (inverse image)
The *inverse image presheaf* $(f^{-1}\mathcal{G})^{\mathrm{pre}}$ is defined by

$$(f^{-1}\mathcal{G})^{\mathrm{pre}}(V) = \frac{\{(s_U, U) \mid f(V) \subseteq U, s_U \in \mathcal{G}(U)\}}{\sim}$$

where $\sim$ denotes sections which agree on a smaller open set containing $f(V)$.

[1] i.e. a surjective morphism of sheaves need not be surjective as a morphism of presheaves...

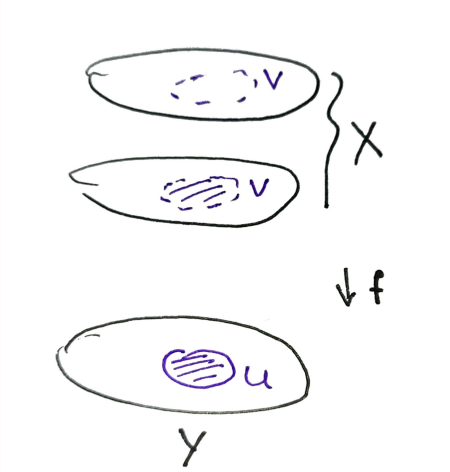


The *inverse image sheaf* is defined by

$$f^{-1}\mathcal{G} = \left((f^{-1}\mathcal{G})^{\text{pre}}\right)^{\text{sh}}$$

Example 2.5.4 (was sheafification necessary?)

Let Y be any topological space, $X = Y \sqcup Y$. Let $\mathcal{G} = \underline{\mathbb{Z}}$ be the consatnt sheaf, and $\mathcal{F} = (f^{-1}\mathcal{G})^{\text{pre}}$.



Fix $U \subseteq Y$ open, and $V = f^{-1}(U)$. By definition, $\mathcal{F}(U) = \mathcal{G}(V) = \mathbb{Z}$ assuming U is connected. But $V = U \sqcup U$, and so

$$\mathcal{F}^{\text{sh}}(V) = \mathcal{G}(U) \times \mathcal{G}(U) = \mathbb{Z}^2$$

Note in this case f is an open map, so the reason for requiring sheafification was not because f does not have to be open.

Example 2.5.5

Let $\mathcal{F}$ be a sheaf on X, $\pi : X \to \{*\}$. Then $\pi_*\mathcal{F}$ is a sheaf on a point. But a sheaf on a point is just an abelian group, in particular, specifically

$$\mathcal{F}(\pi^{-1}(*)) = \mathcal{F}(X)$$

Notation 2.5.6. For a sheaf $\mathcal{F}$ on X, write

$$\mathcal{F}(X) = \Gamma(X, \mathcal{F}) = \mathrm{H}^0(X, \mathcal{F})$$

for the global sections, or the zeroth cohomology with coefficients in $\mathcal{F}$.

For $p \in X$, $i : \{p\} \to X$, $\mathcal{G}$ a sheaf on p, i.e. an abelian group A. Consider $i_*(\mathcal{G})$. The sheaf on X is

$$(i_*\mathcal{G})(U) = \begin{cases} 0 & p \notin U \\ A & p \in U \end{cases}$$

We call this the *skyscraper at p with value A*.

3 Schemes

tl;dr: Spec(A) with a sheaf $\mathcal{O}_{\mathrm{Spec}(A)}$, with $\mathcal{O}_{\mathrm{Spec}(A)}(U_f) = A_f$. Globalise to get a scheme.

Definition 3.0.1 (localisation)
Let A be a ring, $S \subseteq A$ closed under multiplication. The *localisation of A at S* is

$$S^{-1}A = \frac{\{(a, s) \mid a \in A, s \in S\}}{\sim}$$

where $(a, s) \sim (a', s')$ if and only if there exists $s'' \in S$ such that

$$s''(as' - a's) = 0$$

For example, consider $S = \{1, f, f^2, \dots\}$, or $S = A \setminus \mathfrak{p}$ where $\mathfrak{p}$ is a prime ideal.

Remark 3.0.2. Note that the natural map $A \to S^{-1}A$ need not be injective.

What is going to happen? We will define a sheaf $\mathcal{O}_{\mathrm{Spec}(A)}$ on Spec(A), such that

- The stalk at a prime $\mathfrak{p}$ is the localisation $(A \setminus \mathfrak{p})^{-1}A$,
- if U_f is a distinguised open, then $\mathcal{O}_{\mathrm{Spec}(A)}(U_f) = A_f$.

3.1 Sheaf on a base

Fix a topological space X, and $\mathcal{B}$ a basis for the topology.

Definition 3.1.1 (sheaf on a base)
A *sheaf F on the base $\mathcal{B}$* consists of assignments $B_i \mapsto F(B_i)$, with restriction maps $F(B_I) \to F(B_J)$ whenever $B_J \subseteq B_i$, satisfying the usual relations when $B_i \subseteq B_j \subseteq B_k$ and $B_i = B_j$. Moreover, we have the additional axioms:

SB1. If $B = \bigcup_i B_i$, with $B, B_i \in \mathcal{B}$, $f, g \in F(B)$, with $f|_{B_i} = g|_{B_i}$ for all i, then $f = g$.

SB2. If $B = \bigcup_i B_i$, with $B, B_i \in \mathcal{B}$, with $f_i \in F(B_i)$, such that for all $B' \subseteq B_i \cap B_j$, $f_i|_{B'} = f_j|_{B'}$, then there exists $f \in F(B)$ with $f|_{B_i} = f_i$.

Proposition 3.1.2. Let F be a sheaf on a base $\mathcal{B}$ of X. Then this uniquely determines a sheaf $\mathcal{F}$, by $\mathcal{F}(B_i) = F(B_i)$, agreeing with restriction maps.

Proof. We will define the stalks of $\mathcal{F}$ first. Set

$$\mathcal{F}_p = \frac{\{(s_B, B) \mid p \in B \in \mathcal{B}, s_B \in F(B)\}}{\sim}$$

where $(s_B, B) \sim (s_{B'}, B')$ if there exists $B'' \subseteq B \cap B'$ such that $s_B|_{B''} = s_{B'}|_{B''}$.

We can then use the trick from sheafification, that is,

$$\mathcal{F}(U) = \left\{(f_p \in \mathcal{F}_p)_{p \in U} \mid (*)\right\}$$

where the condition $(*)$ is such that: for every $p \in U$, there exists a basic open B containing p, and $s \in F(B)$, with $s_q = f_q$ for all $q \in B$. This is clearly a sheaf.

By the sheaf axioms, the natural maps $F(B) \to \mathcal{F}(B)$ is an isomorphism by the sheaf axioms. □

Lecture 8

Recall $\mathrm{Spec}(A)$ is a topological space, with distinguised open sets $\{U_f\}_{f\in A}$. Moreover, $U_f = U_g$ if and only if f and g are powers of some $h \in A$, and if $U_f = U_g$, then $A_f \cong A_g$. Therefore, the assignment

$$U_f \mapsto A_f$$

is well-defined.

Proposition 3.1.3. The assignment $U_f \to A_f$ defines a sheaf of rings on the base $\{U_f\}$ of $\mathrm{Spec}(A)$.

An immediate consequence is that $\mathrm{Spec}(A)$ inherits a sheaf of rings, denoted $\mathcal{O}_{\mathrm{Spec}(A)}$, called the *structure sheaf*.

Prelude. Suppose $\{U_{f_i}\}_{i\in I}$ covers $\mathrm{Spec}(A)$, then there exists a finite subcover. That is, $\mathrm{Spec}(A)$ is quasicompact. This is on examples sheet 1, but also:

Since the U_{f_i} cover, there is no prime ideal $\mathfrak{p} \trianglelefteq A$ containing all f_i. Equivalently,

$$\sum_{i\in I} \langle f_i \rangle = \langle 1 \rangle$$

But we can write 1 as a finite sum $1 = \sum a_i f_i$. But then if $J \subseteq I$ are the indices with $a_i \neq 0$, then $\{U_{f_i}\}_{i\in J}$ cover. □

Proof of proposition 3.1.3. We need to check SB1 and SB2. We will check these for the basis open $B = \mathrm{Spec}(A)$. The general case is similar (replace A with A_f).

SB1: Suppose we have a cover

$$\mathrm{Spec}(A) = \bigcup_{i=1}^{n} U_{f_i}$$

By the prelude, it suffices to consider the finite case. Given $s \in A$ such that $s|_{U_{f_i}} = 0$ for all i, then by the definition of localisation, we have that $f_i^m s = 0$ for some m large enough. But $\langle 1 \rangle = \langle f_i^m \rangle_{i=1}^n$ for any $m > 0$, as the U_{f_i} cover, which then implies the $U_{f_i^m}$ cover.

With this,

$$s = s \cdot 1 = s \cdot \sum_{i=1}^{n} r_i f_i^m = \sum_{i=1}^{n} r_i s_i f_i^m = 0$$

SB2: Say

$$\mathrm{Spec}(A) = \bigcup_{i\in I} U_{f_i}$$

and choose elements in each A_{f_i}, which agree on $A_{f_i f_j}$. That is, we have $s_i \in A_{f_i}$, with the images of s_i, s_j in $A_{f_i f_j}$ agreeing. We need to build $s \in A$ with these localisations.

First suppose I is finite. On A_{f_i}, we have the element

$$\frac{a_i}{f_i^{\ell_i}} \in A_{f_i}$$

We will write $g_i = f_i^{\ell_i}$, and note that $U_{f_i} = U_{g_i}$.

On overlaps, restrict to $A_{g_i g_j}$. Then we have that

$$(g_i g_j)^{m_{ij}}(a_i g_j - a_j g_i) = 0$$

Rewriting this using algebra and the fact that $U_f = U_{f^k}$ for all k. By taking the largest, we can assume $m = m_{ij}$. We write

$$b_i = a_i g_i^m \qquad h_i = g_i^{m+1}$$

Using this, the element we chose in A_{f_i}, becomes b_i/h_i. But now $U_{f_i} = U_{h_i}$ cover $\mathrm{Spec}(A)$, and so we can write

$$1 = \sum_i r_i h_i$$

Now we construct

$$r = \sum r_i b_i$$

with r_i as above. This restricts correctly to b_i/h_i on U_{h_i} (i.e. in the localisation A_{h_i}).

When I is infinite, choose a finite subcover, or equivalently, $\langle f_1, \ldots, f_n \rangle = A$, and use the above to build $r \in A$. But given $\langle f_1, \ldots, f_n, f_\alpha \rangle = A$, the same construction gives a new element r'. But $r' = r$ by SB1. □

Definition 3.1.4 (structure sheaf)

The *structure sheaf on* $\mathrm{Spec}(A)$ is the sheaf associated to the sheaf on the base sending

$$U_f \mapsto A_f$$

The sheaf is denoted $\mathcal{O}_{\mathrm{Spec}(A)}$.

Remark 3.1.5.

$$\mathcal{O}_{\mathrm{Spec}(A),\mathfrak{p}} = A_{\mathfrak{p}}$$

Definition 3.1.6 (ringed space)

A *ringed space* $(X, \mathcal{O}_X)$ is a topological space X, with a sheaf of rings $\mathcal{O}_X$ on X.

We note that any open subset of a ringed space is naturally a ringed space, with

$$\mathcal{O}_U(V) = \mathcal{O}_X(V)$$

for all $V \subseteq U \subseteq X$ open.

Definition 3.1.7 (isomorphism of ringed spaces)

An *isomorphism of ringed spaces* $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is

- a homeomorphism $\pi : X \to Y$,
- an isomorphism of sheaves on Y, $\mathcal{O}_Y \to \pi_* \mathcal{O}_X$.

Remark 3.1.8. We could have also chosen $\pi^{-1}\mathcal{O}_Y \to \mathcal{O}_X$.

Definition 3.1.9 (affine scheme)

An *affine scheme* is a ringed space $(X, \mathcal{O}_X)$ which is isomorphic to $(\mathrm{Spec}(A), \mathcal{O}_{\mathrm{Spec}(A)})$.

Definition 3.1.10 (scheme)

A *scheme* is a ringed space $(X, \mathcal{O}_X)$, that is locally isomorphic to an affine scheme.

In other words, for every $x \in X$, there exists $x \in U \subseteq X$ open, with $(U, \mathcal{O}_U)$ being an affine scheme.

3.2 Examples of schemes

Example 3.2.1

Spec(A) is a scheme.

Example 3.2.2 (Open subschemes)

Let X be a scheme, $U \subseteq X$ an open subset. We can take

$$\mathcal{O}_U = \mathcal{O}_X|_U$$

to be the structure sheaf of U, making U into a scheme. This is because the $\{U_f\}$s form a basis of the topology on Spec(A).

Example 3.2.3

Take $X = \mathrm{Spec}(\mathbb{C}[x, y])$, and $U = X \setminus \{\langle x, y\rangle\}$. The scheme U is *not* an affine scheme.

Index

affine scheme, 17
affine variety, 3

cokernel
    as a presheaf, 12
    as a sheaf, 12
constant presheaf, 9
constant sheaf, 10
coordinate ring, 3

distinguished open, U_f, 6

germ, 10

image
    as a presheaf, 12
    as a sheaf, 12
injective
    morphism of presheaves, 9
    morphism of sheaves, 13
inverse image
    presheaf, 13
    sheaf, 14
isomorphism
    of affine varieties, 3
    of ringed spaces, 17

kernel
    as a presheaf, 12
    as a sheaf, 12

localisation
    at $f \in A$, 6
    at a subset, 15

morphism
    of affine varieties, 3
    of presheaves, 8
    of sheaves, 9

open subscheme, 18

presheaf of abelian groups, 8
presheaf pushforward, 13

radical, 3
restriction map, 8
ringed space, 17

scheme, 17
section, 10
sheaf, 9
    from a sheaf on a base, 15
sheaf on a base, 15
sheafification, 11
skyscraper sheaf, 15
stalk, 10
structure sheaf
    of Spec(A), 17
    of a variety, 10
surjective
    morphism of presheaves, 9
    morphism of sheaves, 13

Weil conjectures, 2

Zariski spectrum, Spec A, 4
Zariski topology
    of affine varieties, 3
    on Spec A, 5